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ONS Operador Nacional do Sistema Elétrico

Operational Challenges of the Brazilian Interconnected Power System (SIN) in the Era of Distributed Energy Resources

India Smart Utility Week

March 12, 2026

About National Electric System Operator (ONS)

About

Private, non-profit legal entity under the regulation and supervision of ANEEL (Brazilian Electricity Regulatory Agency).

ONS does not own generation, transmission, or distribution assets.

Centralized operation of the Brazilian Interconnected Power System ensures secure and cost-efficient system operation.

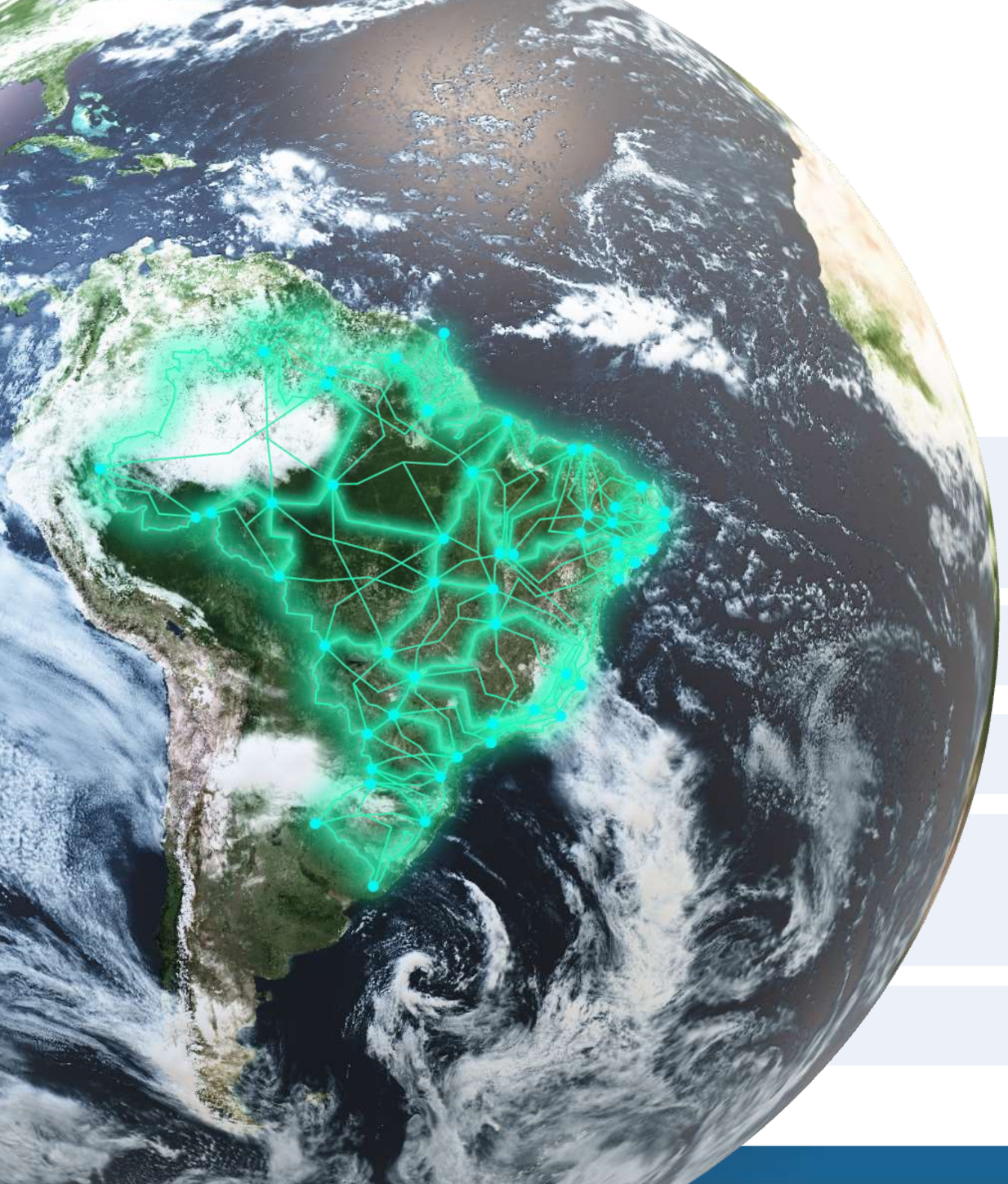
LEGAL STRUCTURE

Article 13 of Law No. 9,648/98

(as amended by Law No. 10,848/04), regulated by Decree No. 5,081/04.

Mission

To guarantee the supply of electricity in the country, with quality and a balance between security and the overall operation cost.



Why Brazil is an Interesting case?

One of the **largest hydro-dominated power** systems in the world;

Continental-scale **interconnected** grid;

Rapid growth of distributed solar generation;

Increasing operational complexity due to DER integration;

Centralized system operation by ONS.

Brazilian Interconnected Power System (SIN)

General Characteristics

Extension of transmission lines ≥ 230 kV (km):

2026: 181.326 km

2029: 196.160 km (expected)

Energy load

Peak: 106,536 MW **26/02/25 at 2:45PM**

Renewables record 26/02/25

The S.I.N covers almost all of Brazil and supplies nearly all of the country's electricity demand.

(90,1 GWmed)

Load supplied by

93.4%

renewable energy

68% 9% 14%

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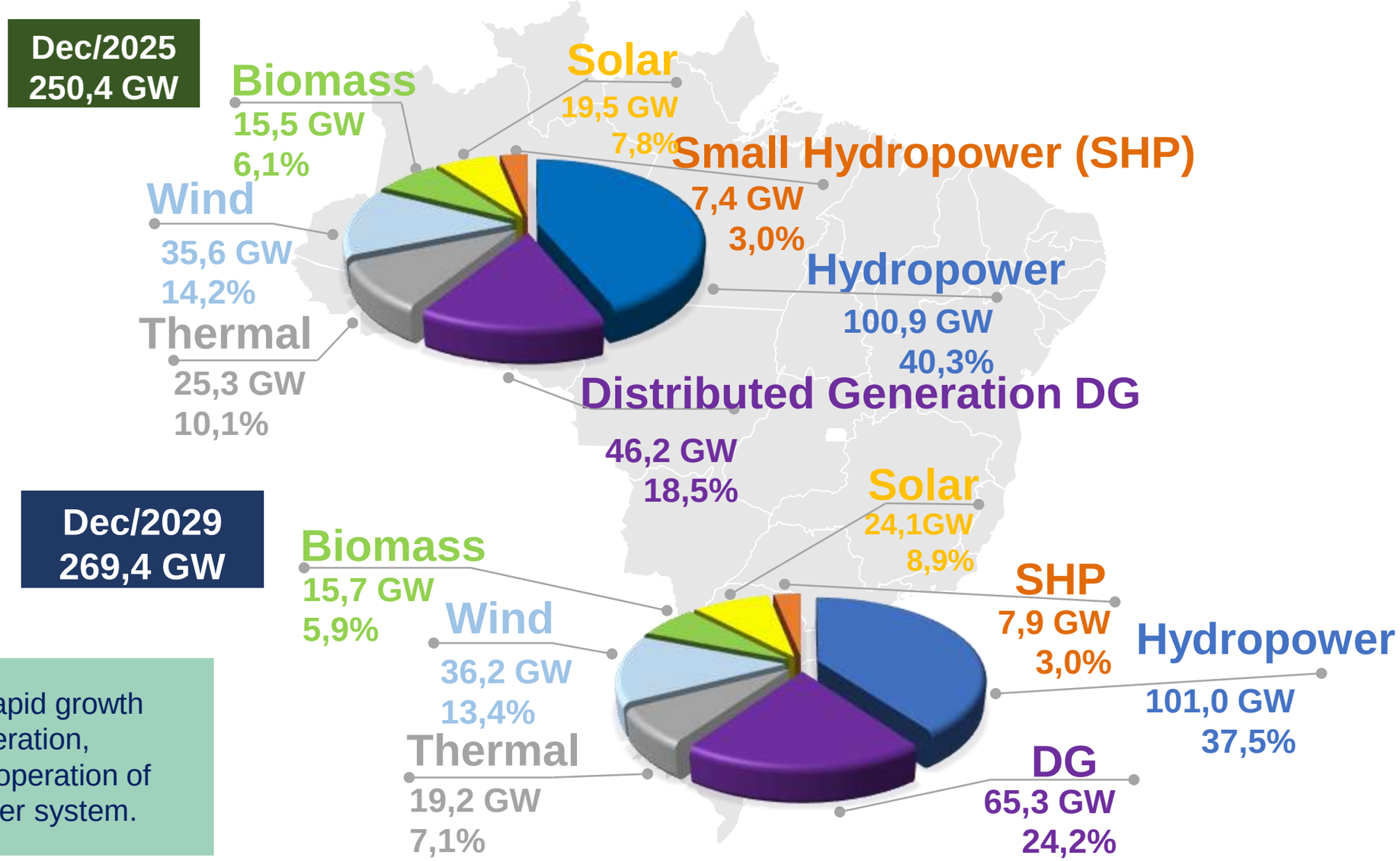
Brazil’s Power Generation Mix: Strong Growth of Renewables and Distributed Generation

Power Generation Mix

Installed Capacity in the SIN – 2026 and 2030



Brazil is experiencing rapid growth in distributed solar generation, which is reshaping the operation of the interconnected power system.

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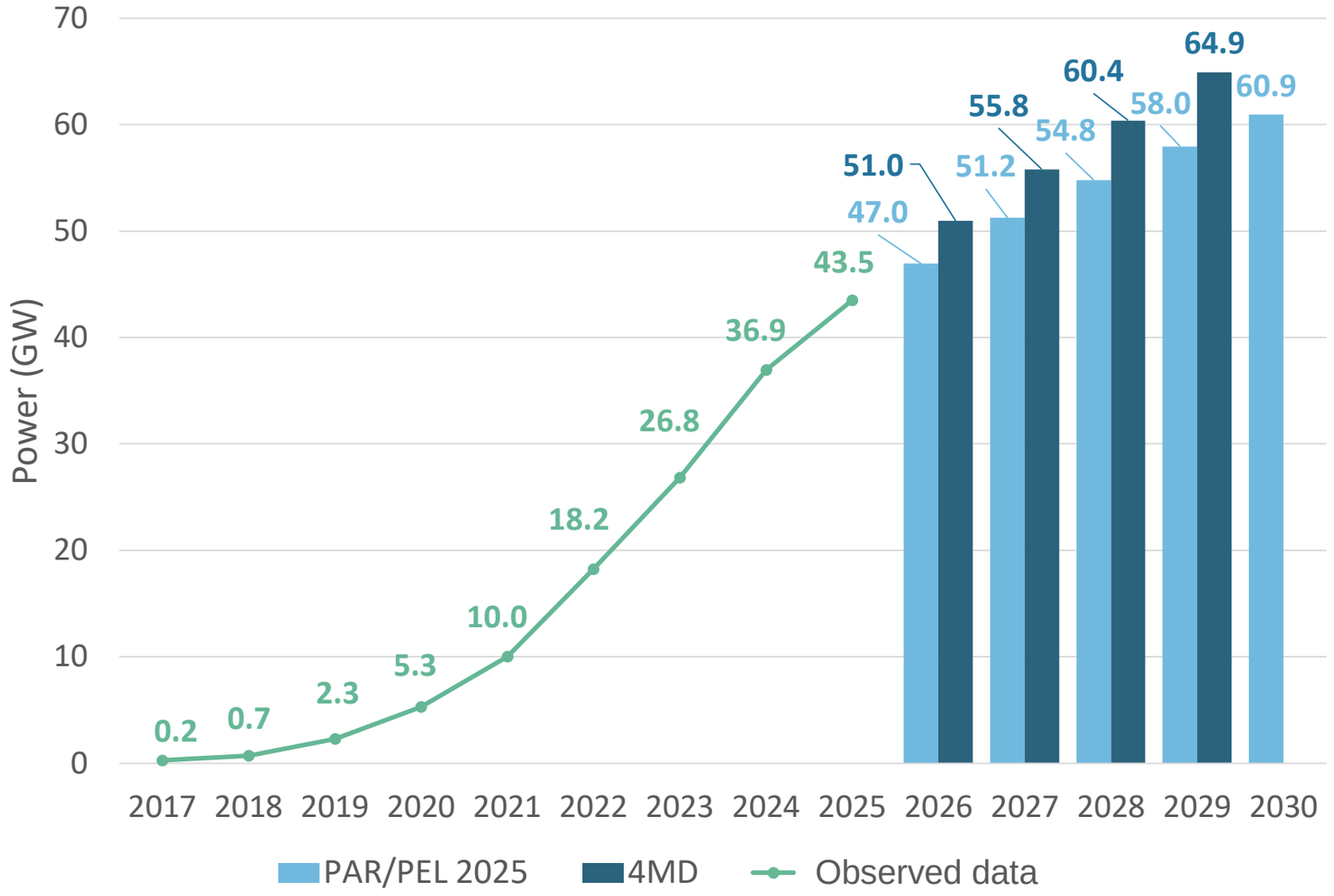
Impacts of Distributed Generation on Power System Operation

Distributed Solar Growth

Rapid Growth of Rooftop Solar in Brazil



Current Situation

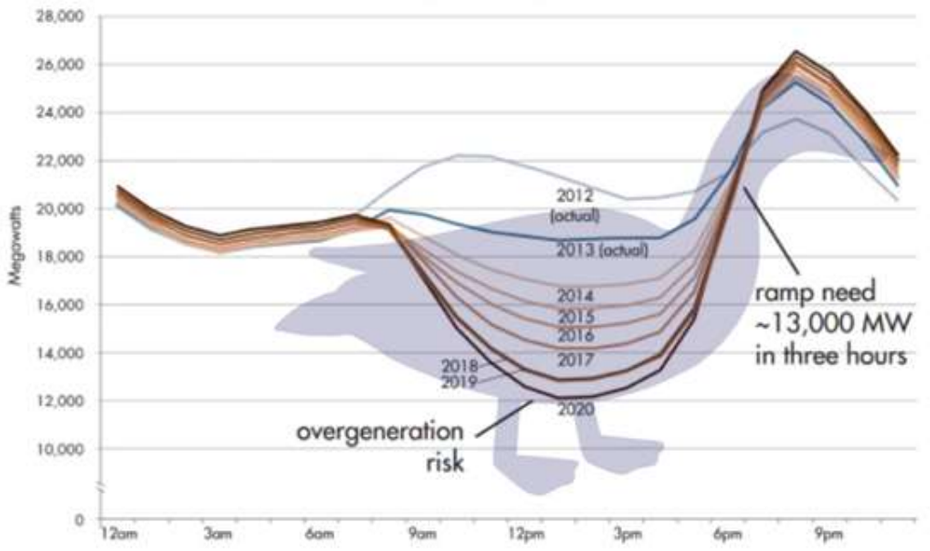
+ ≈ 20 GW of other generation sources connected to distribution networks

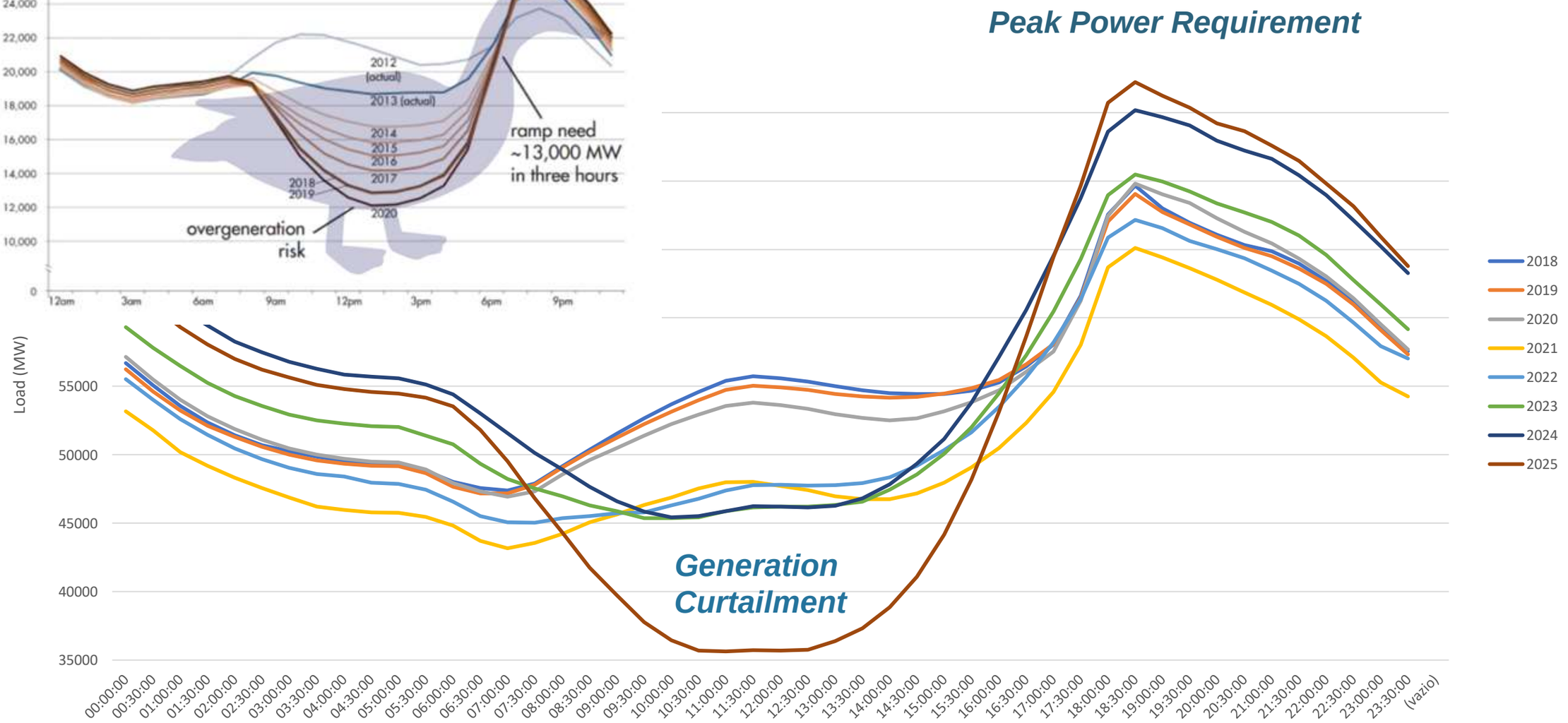
≈ 63,5 GW of generation connected to distribution networks

≈ 25% of the installed capacity of the SIN

SIN Load Curve – Typical Sunday Profiles

August Reference





Impacts of Distributed Generation on Power System Operation

Displacement of generation from large centralized power plants by small-scale resources

Management of constraints and bottlenecks in the transmission system and at the interface with distribution networks

Small-scale resources connected through power-electronic inverters, typically outside the direct control of the system operator

Need for additional resources for automatic voltage control

Bidirectional power flows between transmission and distribution systems

Need for greater operational flexibility, currently mainly provided by hydropower

Need to redefine the role of distribution companies and increase coordination with the ONS

ONS Actions to Address Distributed Generation Impacts

ONS/DSO Interface Project

Creation of a regulatory framework for the management of Distributed Energy Resources (DERs), developed in collaboration with distribution companies.

Parameterization, Validation and Calibration of Load and Distributed Generation Models

Enhancement of load and distributed generation models used in electrical studies.

Review of the Classification of Power Plant Operating Modes

Adjustment of power plant operating modes to the new context of the power sector, characterized by the growing penetration of distributed generation.

Distribution Network Surplus Management Plan

Definition of procedures for reducing generation from GD power plants in emergency situations involving generation surplus in the SIN

Update of Module 3 of PRODIST

Update of technical requirements for the connection of Distributed Energy Resources (DERs) to the distribution network.

Key Conclusions

Key Conclusions

- Distributed generation is already the second largest source of installed capacity in Brazil
- Minimum net load levels are decreasing, increasing the need for operational flexibility and voltage control resources
- Reverse power flows at transmission–distribution interfaces are becoming more frequent, which may cause overloading of transmission network equipment and increase the need for curtailment of centralized generation to ensure operational reliability
- Modernization of technical requirements for DER connection to distribution networks is necessary.
- Improved monitoring and coordination between DSOs and the system operator will be essential.

Additional Challenges for System Operation

- Greater availability of firm capacity to meet peak demand
- Adequate dynamic modeling capability for all power-electronics-based equipment connected to the grid (generation and load).
- Greater availability of additional voltage control resources and increase of system short-circuit strength.
- Regular capacity auctions to ensure sufficient resources to meet system peak demand.
- Regulatory framework for Battery Energy Storage Systems (BESS) and rules for large consumers to access the SIN.
- Reassessment of public policies related to subsidies and incentives for renewable energy, aligned with the current sector context.;
- Discussion on the inclusion of part of the Distributed Energy Resources (DERs) in the commercial and physical allocation of generation curtailment.;

OUR CHANNELS

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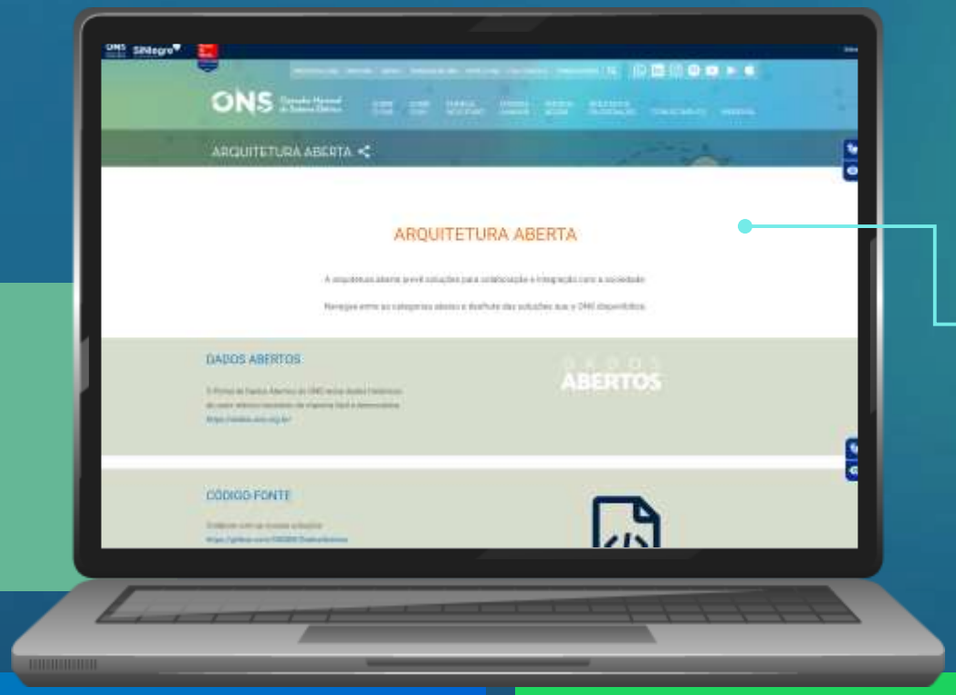

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Thank you!

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